Лабораторная работа № 6. Статическая маршрутизация VLAN

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Информация

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Цель работы

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Настроить статическую маршрутизацию VLAN в сети.

Задание

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1. Добавить в локальную сеть маршрутизатор, провести его первоначальную настройку.
2. Настроить статическую маршрутизацию VLAN.
3. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

Настройка маршрутизатора

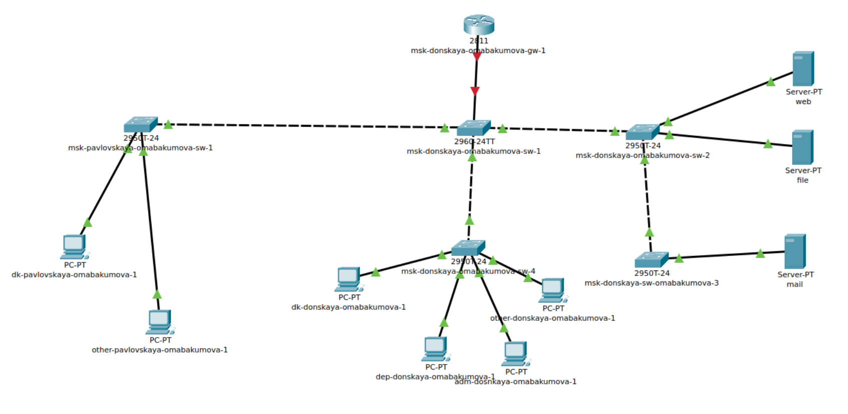


Рис. 1: Размещение маршрутизатора

Настройка маршрутизатора

```
Router(config)#hostname msk-donskaya-omabakumova-gw-1
msk-donskaya-omabakumova-gw-1(config)#line vty 0 4
msk-donskaya-omabakumova-gw-1(config-line)#password cisco
msk-donskaya-omabakumova-gw-1(config-line)#login
msk-donskaya-omabakumova-gw-1(config-line)#exit
msk-donskaya-omabakumova-gw-1(config)#line console 0
msk-donskaya-omabakumova-gw-1(config-line)#password cisco
msk-donskaya-omabakumova-gw-1(config-line)#login
```

Рис. 2: Первоначальная настройка маршрутизатора

Настройка маршрутизатора

```
msk-donskaya-omabakumova-gw-1(config)#enable secret cisco
msk-donskaya-omabakumova-gw-1(config)#service password-encryption
msk-donskaya-omabakumova-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-omabakumova-gw-1(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-omabakumova-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-omabakumova-gw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-omabakumova-gw-1(config)#line vty 0 4
*Mar 1 0:8:20.391: %SSH-5-ENABLED: SSH 1.99 has been enabled
msk-donskaya-omabakumova-gw-1(config-line)#transport input ssh
msk-donskaya-omabakumova-gw-1(config-line)#exit
msk-donskaya-omabakumova-gw-1(config)#exit
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 3: Первоначальная настройка маршрутизатора

Настройка маршрутизатора

```
User Access Verification

Password:

msk-donskaya-omabakumova-sw-1>enable
Password:
msk-donskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-omabakumova-sw-1(config)#interface f0/24
msk-donskaya-omabakumova-sw-1(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-1(config-if)#^Z
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-1#
```

Рис. 4: Настройка порта 24

Настройка статической маршрутизации

```
msk-donskaya-omabakumova-gw-1>enable
Password:
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#interface f0/0
msk-donskaya-omabakumova-gw-1(config-if)#no shutdown

msk-donskaya-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-omabakumova-gw-1(config-if)#exit
msk-donskaya-omabakumova-gw-1(config)#interface f0/0.2
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 2
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.1.1 255.255.255.0
msk-donskaya-omabakumova-gw-1(config-subif)#description management
msk-donskaya-omabakumova-gw-1(config-subif)#exit
msk-donskaya-omabakumova-gw-1(config)#inteface f0/0.3
                                         ^
% Invalid input detected at '^' marker.

msk-donskaya-omabakumova-gw-1(config)#interface f0/0.3
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 3
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-omabakumova-gw-1(config-subif)#description servers
```

Рис. 5: Конфигурация VLAN-интерфейсов маршрутизатора

Настройка статической маршрутизации

```
msk-donskaya-omabakumova-gw-1(config-subif)#interface f0/0.101
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.101, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 101
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-omabakumova-gw-1(config-subif)#description dk
msk-donskaya-omabakumova-gw-1(config-subif)#interface f0/0.102
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.102, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 102
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
msk-donskaya-omabakumova-gw-1(config-subif)#description departments
msk-donskaya-omabakumova-gw-1(config-subif)#interface f0/0.103
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.103, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 103
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
msk-donskaya-omabakumova-gw-1(config-subif)#description adm
msk-donskaya-omabakumova-gw-1(config-subif)#interface f0/0.104
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.104, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 104
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
msk-donskaya-omabakumova-gw-1(config-subif)#description other
```

Рис. 6: Конфигурация VLAN-интерфейсов маршрутизатора

Настройка статической маршрутизации

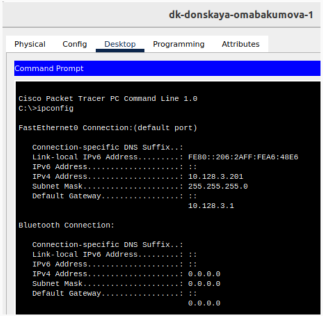
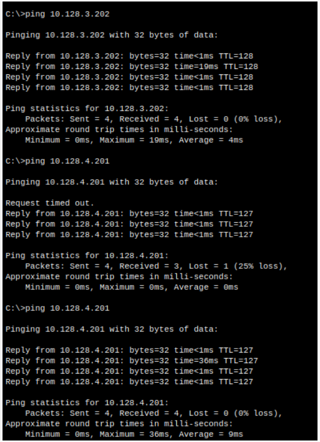


Рис. 7: Вывод информации об оконечном устройстве

Настройка статической маршрутизации



```
C:\>ping 10.128.3.202

Pinging 10.128.3.202 with 32 bytes of data:

Reply from 10.128.3.202: bytes=32 time<1ms TTL=128
Reply from 10.128.3.202: bytes=32 time=19ms TTL=128
Reply from 10.128.3.202: bytes=32 time<1ms TTL=128
Reply from 10.128.3.202: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.202:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 19ms, Average = 4ms

C:\>ping 10.128.4.201

Pinging 10.128.4.201 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.201: bytes=32 time<1ms TTL=127
Reply from 10.128.4.201: bytes=32 time<1ms TTL=127
Reply from 10.128.4.201: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.201:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.4.201

Pinging 10.128.4.201 with 32 bytes of data:

Reply from 10.128.4.201: bytes=32 time<1ms TTL=127
Reply from 10.128.4.201: bytes=32 time=36ms TTL=127
Reply from 10.128.4.201: bytes=32 time<1ms TTL=127
Reply from 10.128.4.201: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 36ms, Average = 9ms
```

Рис. 8: Проверка доступности с помощью ping

Настройка статической маршрутизации

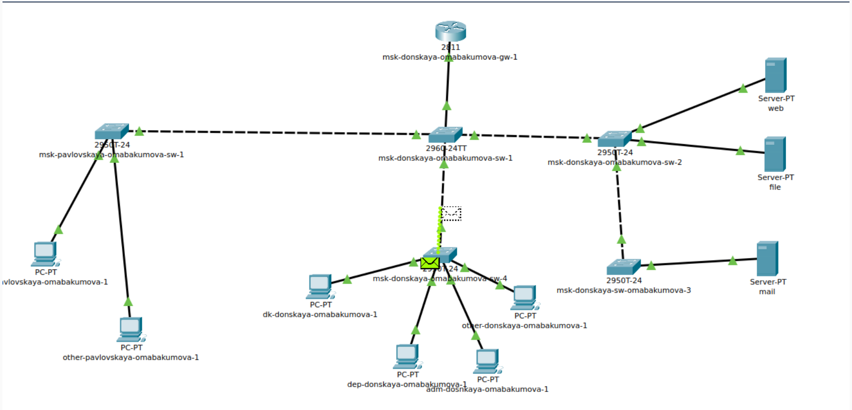


Рис. 9: Передвижение пакетов по сети

Настройка статической маршрутизации

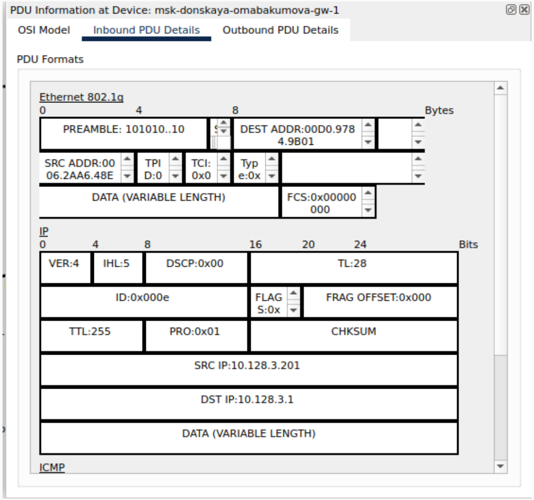


Рис. 10: Информация о PDU

Выводы

Выводы

Во время выполнения данной лабораторной работы я приобрела настраивания статической маршрутизации VLAN в сети.